PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc (Food Processing and Preservation)

as per NEP 2020

w.e.f session 2026-27

Course Outline for B.Voc (Food Processing and Preservation) as per NEP 2020 w.e.f. session 2026-27

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 101	Core Discipline	Fundamentals of Food Science	2	2	4	40	10	40	10	100
BFP 102	Core Discipline	Bakery and Confectionery Technology	2	2	4	40	10	40	10	100
BFP 103	Core Discipline	Quality Assurance	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ex (Pr)	Int (Pr)	Total
BFP 201	Core Discipline	Food Preservation Technology	2	2	4	40	10	40	10	100
BFP 202	Core Discipline	Food Plant Layout	2	2	4	40	10	40	10	100
BFP 203	Core Discipline	Dairy Science and Technology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Bakery Technician

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Food Processing and Preservation = 48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 301	Core Discipline	Nutrition and Food Chemistry	2	2	4	40	10	40	10	100
BFP 302	Core Discipline	General Microbiology	2	2	4	40	10	40	10	100
BFP 303	Core Discipline	Foods Of Plant Origin	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC III	Skill Enhancement Course	SEC Options[1]			3					75
B.Voc. MDSC 301	Multidisciplinary Course	Entrepreneurship Development[1]	3	-	3	60	15	-	-	75
VAC II	Value-Added Course	VAC Options[3]			2					50
PW II	Project Work	Project Work-II & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 401	Core Discipline	Food Packaging Technology	2	2	4	40	10	40	10	100
BFP 402	Core Discipline	Food Microbiology	2	2	4	40	10	40	10	100
BFP 403	Core Discipline	Foods Of Animal Origin	2	2	4	40	10	40	10	100
BFP 404	Core Discipline	Documentation And Record Keeping	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
VAC III	Value-Added Course	VAC Options[3]			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Food Microbiologist

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 5th Semester. Viva-Voce examination to be held within 3 weeks of the start of 5 th semester.

** Minimum Credits Required to get Diploma Course in Food Processing and Preservation = 96 + 4 #.

#Exit 2: Diploma Course - Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the second year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 501	Core Discipline	Food Analysis	2	2	4	40	10	40	10	100
BFP 502	Core Discipline	Industrial Safety	2	2	4	40	10	40	10	100
BFP 503	Core Discipline	Principles of Management in Food Industry	4	-	4	80	20	-	-	100
BFP 504	Core Discipline	Technology Of Fruits And Vegetables	2	2	4	40	10	40	10	100
BFP 505	Core Discipline	Technology Of Plantation Crops & Oilseeds	2	2	4	40	10	40	10	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work-III & Viva-Voce			2					50
Total Credits					**24**					**600**

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 601	Core Discipline	Bioanalytical Techniques	2	2	4	40	10	40	10	100
BFP 602	Core Discipline	Food Safety, Standards And Regulation	2	2	4	40	10	40	10	100
BFP 603	Core Discipline	Food Pathogens	2	2	4	40	10	40	10	100
BFP 604	Core Discipline	Waste Management and Environmental Control in Food Processing	2	2	4	40	10	40	10	100
BFP 605	Core Discipline	Meat & Seafood Processing	2	2	4	40	10	40	10	100
BFP 606	Core Discipline	Egg & Poultry Processing	2	2	4	40	10	40	10	100
Total Credits					**24**					**600**

Job Role - Food Safety Analyst

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Skill Enhancement Courses (SEC): Annexure I
2. Multidisciplinary Courses (MDSC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF B.VOC
(Food Processing and Preservation)
(SEMESTER 1 to 6)

PROGRAMME OUTCOMES

PO1. Understanding of the concepts, principles of working of the Food Industry.

PO2. To understand the different preservation techniques of foods by heat processing, by low temperature, by removal of water and by using salt and sugar or by other relevant methods._.

PO3. To identify the various microorganisms pathogenic or non-pathogenic and trace the cause of food contamination.

PO4. Describe the spoilage and deterioration mechanisms happening in foods and food products and to apply methods to control microbial activities prolonging shelf life, assuring it fit for human consumption.

PO5. To plan and design a Food Plant Layout in accordance with the type of establishment (small, medium or large) in a specific area.

PO6. To understand Food laws and Food Safety regulations in India and worldwide, this will be accompanied by Safety and hygienic practices to be followed in a food plant layout.

PO7. To develop ability to apply principles of food engineering in industry,

PO8. Process and prepare food products which could be brought in the concurrent food market for sale.

PO9. To develop awareness of the marketing strategies, management hierarchy involved in a food processing plant.

Semester I

<table>
<tr><td>1</td><td>Course Title</td><td>Fundamentals of Food Science</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 101</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand the importance of nutrients in our daily diet and to formulate nutritionally enriched food products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Develop proficiency skill in producing different nutritious food products. .
CO2. Make different processed food products with quality assurance. .
CO3.Understand the structure, composition, and nutritional quality of foods.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Introduction to Food Science: Definition & Scope. Basic concept of Food: Nutrient, Nutrition, Classification of Nutrients : Macronutrients, Micronutrients; dietary guidelines. Concept of food groups. **UNIT– II** Constituents of food - Definition, occurrence, properties, significance : Carbohydrates, Proteins, Fats and oils; Nutritive aspects. Protein quality. Bioavailability of nutrients. Vitamins. Minerals. Fiber. Water. **UNIT– III** Food and nutrition - Energy metabolism - BMR. Recommended dietary allowances, Balanced diet for different age groups (Infancy to old age). Stability of nutrients. Role of nutrients in health and disease prevention. **UNIT-IV** Food Processing: Unit operations in food industries : Materials handling, cleaning, separating, grinding, mixing, heat exchanging, forming, packaging. **Practical Work** 1. Demonstration of basic laboratory equipments used in food processing labs. 2. Demonstration of chemistry of candy making and the effect of temperature on the texture of candies. 3. Calculation of BMR 4. Enrichment and fortification of routine diet. 5. Demonstration of unit operation in food processing :mixing, separating, heat exchanging, packaging.
12	**Suggested Resources for Reading** 1. Food Science by B. Srilalshmi, New Age International Private Limited; Sixth edition, 2015. 2.Food: Facts and Principles by N. Shakuntala Manay, M. Shadaksharaswamy, New Age International Private Limited; Fourth edition, 2020. 3. Food Science by Joseph H. Hotchkiss and Norman N Potter,Aspen Publishers Inc.,U.S, Fifth edition,2007 4. Essentials of Food Science by Vickie A. Vaclavik, Elizabeth W. Christian, Springer Publications,Fourth edition,2013 5. Food Science by Sunetra Roday., Oxford publication, Third edition, 2018

<table>
<tr><td>1</td><td>Course Title</td><td>Bakery and Confectionary Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 102</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
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</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

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iv. The students are required to attempt one long question of 8 marks from each unit.
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<tr><td>7</td><td>Job Role</td><td>Bakery technician.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To impart basic knowledge of scope of bakery and confectionery and learn about baking process.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Familiarizing the students with different equipment and ingredients used in bakery unit and their functions..
CO2. The students will be able to identify various bakery and confectionery products
CO3.The learners will develop fundamental baking techniques.
CO4. Work effectively as a member of a team and learn entrepreneurial skills for self reliance.
CO5. The students will learn the basics of setting up a bakery unit.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** **Overview of bakery and confectionery industries:** Wheat grain structure and flour milling technology Flour types and quality parameters (protein, ash, WAP, gluten) Functional role of ingredients: Flour, sugar, fats, eggs, milk, salt **Dough and batter systems:** Dough rheology and gluten development. Batter aeration and viscosity **Leavening systems:** Biological (yeast), chemical (baking powder and baking soda) and mechanical leavening **UNIT– II** **Bread Processing:** Straight dough, sponge dough, no-time dough methods, dough mixing. Fermentation, proofing, baking and cooling reactions. **Biscuit and Cookie Processing:** Types of biscuits and cookies, Dough mixing, sheeting, moulding and baking Effect of temperature and time on biscuit quality. **Cake Processing:** Cake structure and foam formation. Creaming, foaming, sugar batter and flour batter methods. Types of cakes and quality parameters. **UNIT– III** **Sensory attributes of bakery products:** Appearance, texture, flavor, aroma and taste Factors affecting quality of bakery products: Flour quality, Ingredient balance, Processing conditions **Bread faults and remedies:** Poor loaf volume, Dense crumb, Cracked crust, Sour flavor Shelf-life and spoilage of bakery products **Bakery equipment:** Small-scale equipment: mixers, ovens, proofers, moulders **Large-scale industrial equipment:** continuous mixers, tunnel ovens **UNIT-IV** **Types of confectionery:** Sugar confectionery, Chocolate confectionery. Raw materials used in confectionery: Sugar, glucose syrup, fats, cocoa, milk solids, emulsifiers Processing of sugar confectionery: Toffee and caramel manufacture, Fondant preparation and applications Hard-boiled sweets: principles and processing steps Sugar-panned confectionery: Principles of panning, Types: hard panning and soft panning Quality defects in confectionery products Packaging and storage of confectionery items **Practical work** 1. Preparation of White Bread. 2. Preparation of Fruit cake. 3. Studying the effect of temperature on the process of biscuit making. 4. Preparation of Garlic bread and sensory analysis. 5. Preparation of Icing of Cakes. 6. To estimate WAP of the flour sample 7. To determine gluten content in the dough
12	**Suggested Resources for Reading** 1. Textbook of bakery and confectionery, by Yogambal Ashok Kumar 2. Bakery products science and technology,by Weibiao zhou, Wiley blackwell, second edition 3. A professional Text to bakery and confectionary by John Kings lee 4. Bakery-I: Student handbook and practical manual- CBSE. 5. Bread: A baker's book of techniques and recipes by Jeffrey Hamelman-Wiley, 2012. 6.Confectionery Science and Technology by _Richard W. Hartel · Joachim H. von Elbe Randy Hofberger,Springer ,2017

<table>
<tr><th>1</th><th>Course Title</th><th>Quality Assurance</th></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 103</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
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(1 Semester = 15 weeks)
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

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ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

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iv. The students are required to attempt one long question of 8 marks from each unit.
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<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To develop a broad understanding of the principles of modern quality management systems; to understand objective and subjective factors that contribute to the quality of foods or food-related services within the food value chain.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Illustrate the importance of food quality standards and of Quality Assurance.
CO2.Understand the principles of Total Quality Management to evaluate and select between alternative strategies in designing a quality management system.
CO3.Explain the application of food quality and food safety system.
CO4.Understand the role of food quality assurance system standards in India.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
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11	**Course Content** **UNIT– I** Concept of quality: quality control vs quality assurance. Importance of quality assurance in food processing. Functions of quality control department in food industries Quality attributes of food: physical, chemical, microbiological and sensory Quality assessment of: Raw materials, In-process products, Finished food products Role of quality assurance in shelf-life, consumer safety and brand value **UNIT– II** Total quality management: Principles and benefits..Quality management tools: quality circles, continuous improvement. Good Manufacturing Practices (GMP) in food processing plants. Good Hygienic Practices (GHP) and sanitation control. Good Laboratory Practices (GLP) in food testing laboratories. Role of personnel training and management commitment. **UNIT– III** Overview of food laws and regulations in India: Food Safety and Standards Act (FSSA), 2006: Objectives and salient provisions. Food standards and regulatory bodies: FSSAI, AGMARK, BIS International standards and systems: Codex Alimentarius, HACCP principles and implementation, ISO 9001 and ISO 22000 (overview), USFDA (basic concept). Importance of food safety management systems in global trade **UNIT-IV** Documentation and record keeping under quality systems: Types of documents: SOPs, work instructions, specifications, batch records. Product Recall and Crisis Management: Types of product recalls and their significance. Recall procedures and responsibilities. Types of audits: internal, second-party, third-party audits. Role of audits in continuous improvement **Practical work** 1.Techniques of quality assessment of different natural and processed foods. 2 Identification and ranking of food product attributes 3. Study of cleaning and sanitizers used in pre and post-operative processes in food industry 4. Preparation of QA documents including raw material specifications and process flow charts. 5. Conduct of a mock internal audit: preparation of audit checklist, identification of non-conformities, and preparation of audit report.
12	**Suggested Resources for Reading** 1. Food Quality Assurance Principles and Practices by InteazAlli, CRC Press, 2004. 2. Quality Assurance for the Food Industry. A Practical Approach by J Andres Vasconcellos , CRC Press, 2003 3. Safety And Quality Assurance In Food Supply Chain: Emerging Technologies and challenges by K.P. Sudheer, Bindu Lakshmanan New India Publishing Agency (2021) 4. Food Science by Sunetra Roday., Oxford publication, Third edition, 2018 5. Sensory Evaluation Techniques by Morten C. Meilgaard, B. Thomas Carr, Gail Vance Civille ,CRC Press LLC, Fourth edition, 2006. 6. Principles of Sensory Evaluation of food by Amerine, Pangborn & Roessler , Academic Press, London.2013

Semester II

<table>
<tr><td>1</td><td>Course Title</td><td>Food Preservation Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 201</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
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(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

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<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>The goal of this course is to provide students with a fundamental understanding of food preservation</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Will be able to comprehend the reason for food spoilage.
CO2. Will be able to comprehend the principles of preventing contamination and the removal of microorganisms methods used for food safety.
CO3. Students will be able to understand various heat preservation techniques, low temperature techniques.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Scope and Benefits of food preservation to food industry and consumers. Classification of foods: Perishable Semi-perishable, Non-perishable Causes of food spoilage: Microbial, Enzymatic, Chemical and Physical factors Preservation by salt and sugar: Principle of osmotic pressure and water activity Methods of salting and sugaring. **UNIT– II** Thermal Processing Methods: Principles of thermal preservation, Blanching: Objectives, methods and equipment. Pasteurization: Types (LTLT, HTST, UHT) and applications. Canning: Principles, process steps and canning defects. Sterilization: Commercial sterilization and retort processing Evaporation and concentration: Principles and industrial evaporators Preservation by Fermentation: Definition and importance of fermentation. Advantages and limitations of fermentation as preservation technology. **UNIT– III** Preservation by Low Temperature: Principles of chilling and freezing. Methods of refrigeration and freezing. Freezing equipment and cold storage. Effect of low temperature on food quality and shelf life Preservation by Drying and Dehydration: Principles of drying and moisture removal, Sun drying and mechanical drying Dehydration methods: Tray drying, Spray drying, Freeze drying Concentration of liquid foods. Effect of drying and concentration on nutritional and sensory quality **UNIT-IV** Advanced preservation techniques: Pulsed Electric Field (PEF) Processing: Principle, applications and advantages. High Pressure Processing (HPP): Principle and effect on microorganisms and enzymes Ultrasound Processing: Principles and food applications Microwave Heating: Principle and industrial uses Ohmic Heating:Principle and benefits Infrared Heating: Applications in food processing Advantages and limitations of emerging technologies. **Practical work** 1. Demonstration of various machinery used in food processing. 2. Demonstration on the effect of blanching on the quality of foods. 3. Preservation of food by high concentration of sugar i.e. preparation of jam 4. Preservation of food by using salt 5. Preservation of milk by condensation/concentration 6. To Demonstrate the Preservation of Fruits by drying
12	**Suggested Resources for Reading** 1. Food Science- Potter, CBS publishers. 2. Food Processing and Preservation- Subbulaksmi G., and Udipi S 3. Principles of Food Science, Vol. II- G. Borgstron, Mc. Millan Co. Ltd. London 4. _Principles of food preservation Part I& II- Owen R. Fenemma. 5. Technology of Food Preservation - N.W. Desroiser and N.W. Desrosier

<table>
<tr><td>1</td><td>Course Title</td><td>Food Plant Layout</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 202</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>• Exposure of the students to the basic setup of a Food industry
• To make them conversant with the machinery and equipment in a food industry.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. understand concepts of plant layout.
CO2. have knowledge on building, utilities in the plant.
CO3. know the importance of proper food plant design and safety.
CO4. evaluate and develop problem solving skills regarding food processing operations that can affect the quality of foods.
CO5. know and implement good manufacturing practice.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Industrial plant layout and design: Basic concepts of Plant Layout with reference to food industries. Hygiene requirements in food plant layout and design. **UNIT– II** Plant Design: Design consideration for location of food plants. Basic understanding of equipment layout and ventilation in food processing plants. Preparation of flow sheets for material movement and utility consumption in food plants. **UNIT– III** Plant Layout : Preparation of a Plant Layout, Plant Layout problem, importance, objectives, classical types of layouts. Evaluation of layout. Advantages of good layout **UNIT-IV** Plant Layout Requirements : Common materials of construction of Food plant, building. Maintenance of Food Plant Building, Illumination and ventilation, Cleaning and sanitization, painting and color coding, Fly and insect control. **Practical Work** 1. Study of different types of plant layouts (Dairy, bakery, fruit processing , beverage plant) 2. Preparation of process flow diagrams (Milk Processing plant, fruit juice plant, bakery unit) 3. Study of building materials used in food plants (Floors, ceilings, corrosion resistant, food grade materials, cleaning and maintenance aspects) 4. Study of ventilation, lighting and drainage system of food plant. 5. Study safety and regulatory compliance for food plant design (FSSAI, GMP, GHP, Fire exits, emergency routes) 6. Mini Project (Group/Individual) on food plant layout (Designing layout for food processing unit) 7. Industrial visit/Virtual tour and report making.
12	**Suggested Resources for Reading** 1. Hygienic Design of Food Factories, by John Holah, H. L. M. Lelieveld, Elsevier Publication, 2011 2.Practical Design, Construction and Operation of Food Facilities by J. Peter Clark, Academic Press Publishers.2008 3. Food Plant Economics by Zacharias B. Maroulis, George D. Saravacos ,CRC Press Publishers.2007 4., Food Plant Design by Antonio Lopez-Gomez, Gustavo V. Barbosa-Canovas, CRC Press Publishers.First edition, 2005

<table>
<tr><td>1</td><td>Course Title</td><td>Dairy Science and Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 203</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
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<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months be
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The cal
provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Q
(COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students a
out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Students will learn about _Principles of Dairy, processing of technologies to handle physical aspects of fluid based milk and its various products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. In-depth knowledge of dairy production and processing techniques.
CO2.Comprehensive understanding of dairy production, processing, and preservation techniques.
CO3.Better understanding of the concepts underlying Dairy processing as well as the reliability and caliber of Dairy products.
CO4. Students will learn about technologies to handle physical aspects of fluid based process
CO5.Students will learn about various equipments present in advanced dairy</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Introduction: Milk definition, composition of milk, nutritive value of milk, physico-chemical properties of milk constituents and physicochemical properties of milk, factors affecting quality of milk. **UNIT– II** Market Milk: Manufacture, packaging and storage of standardized pasteurized milk, Various methods of pasteurization, Judging and grading of milk, Flavour defects in milk, their causes and Prevention, **UNIT– III** Special milks: Sterilized milk, Homogenized milk, Flavoured milks, Reconstituted/Rehydrated milk, Recombined milk, Toned milk, Double Toned milk Fermented milk: Acidophilus milk, bulgarian buttermilk, Kumiss and kefir. **UNIT-IV** Dairy Products : Composition and manufacturing of butter, butteroil, Ice cream and cheese Practical work 1. To perform platform tests for milk analysis- analysis of milk for its organoleptic properties 2. To determine stability of milk using COB Test 3. To test milk quality by performing MBRT Test 4. To perform Alcohol test to assess raw milk quality 5. Sediment test to check for adulteration in milk sample 6. Determination of titratable acidity of milk 7. To determine Fat content by Gerber method
12	**Suggested Resources for Reading** 1.Outlines of Dairy Technology, by Sukumar De, Oxford University Press. 2. Milk Hygiene In Milk Production Processing and Distribution, F.A.O Publication. Note: Latest edition of text books may be used.

Semester III

<table>
<tr><td>1</td><td>Course Title</td><td>Nutrition and Food Chemistry</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 301</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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<tr><td>6</td><td>Max. Marks and Time</td><td>
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<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand the importance of nutrients in our daily diet and to formulate nutritionally enriched food products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Describe details of the physical and chemical interactions between food components and their impact on quality.
CO2.Understand the structure, composition, and nutritional quality of foods.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Carbohydrates- Structure and Classification of important carbohydrates, glycoproteins and glycolipids. Food sources of carbohydrates. Dietary fibre. **UNIT– II** Proteins- Structure and classification of amino acids, Food sources of proteins. Enzymes- Classification & Nomenclature. Coenzymes, Factors affecting enzyme efficiency. Enzyme kinetics. **UNIT– III** Lipids- Structure & Classification, saturated, unsaturated and phospholipids. Physical properties- melting point, smoke, flash and fire point. Chemical properties- reichert meissel value, polenske value, iodine value, peroxide value, saponification value. **UNIT-IV** Changes in food during processing- Protein denaturation, Oxidative and Hydrolytic Rancidity. Concept of Gelatinisation and retrogradation of starches, Effect of frying on fats. **Practical Work** Estimation of minerals. Determination of thermal inactivation time of enzymes in fruits and vegetables. Estimation of iodine value. Estimation of peroxide value. Protein estimation in food items. Carbohydrate estimation in food items.
12	**Suggested Resources for Reading** 1. Aurand, L. W. and Wood, A. E. (1973). Food Chemistry. The AVI Publishing Co., Connecticut. 2. Belitz, H. D., Grosch, W. and Schieberler, P. (2004). Food Chemistry. Springer, Berlin. 3. DeMan, J. M. (1999). Principles of Food Chemistry. A Chapman and Hall Food Science Book, Aspen Publ., Inc., Gaithersburg, Maryland. 4. Fennama, O. R. (ed). (1996). Food Chemistry. Marcel Dekker, Inc., New York.

<table>
<tr><td>1</td><td>Course</td><td>General Microbiology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 302</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
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iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand microorganisms and the concept of cleanliness, sterilization, maintenance of lab equipment, microbial growth, microscopy and culture medium.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Gain knowledge about the different cell organelles of microorganisms and their detailed functions.
CO2. Study the growth and control of microbes
CO3. Learn about the different types of microorganisms and their significance.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** History of Development of Microbiology: Spontaneous generation vs. biogenesis. Contributions of Antonie von Leeuwenhoek, Louis Pasteur, Robert Koch, Joseph Lister, Alexander Fleming. Characteristics of Predominant microbes in food: Bacteria, fungi, yeast, **UNIT– II** Structure of prokaryotic cells, cell wall structure and composition of Gram Positive and Gram negative bacteria Difference between prokaryotic and eukaryotic cell. **UNIT– III** Concept of sterilization, physical methods of sterilization (moist heat and dry heat sterilization, ionizing and non-ionizing radiations, filtration), chemical methods of sterilization (alcohols, phenols, ethylene oxide) **UNIT-IV** Microbial growth: Phases of microbial growth, growth curve, generation time, specific growth rate, factors affecting optimum microbial growth. Chemostat and turbidostat. Culture medium: Concept and principles of culture media, Classification and types based on: consistency (liquid, semi-solid and solid), Constituents/ ingredients (Complex, defined, selective, differential and enrichment media) **Practical work** Study of basic equipment: laminar flow, homogenizer, autoclave, biosafety systems, light microscope Sterilization of glassware and other lab equipment Preparation and sterilization of liquid and solid media for isolation of microbes Inoculation and other aseptic procedure Simple staining, negative staining and Gram's staining.
12	**Suggested Resources for Reading** Pelczar MJ, Chan ECS and Krieg NR. Microbiology. McGraw Hill Book Company. Stanier RY, Ingraham JL, Wheelis ML, and Painter PR. General Microbiology. McMillan. Tortora GJ, Funke BR and Case CL. Microbiology: An Introduction. Pearson Education Willey, J. M., Sherwood, L., & Woolverton, C. J. (2011). Prescott's microbiology (Vol. 7). New York: McGraw-Hill.

<table>
<tr><td>1</td><td>Course Title</td><td>Foods Of Plant Origin</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 303</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
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</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To develop a broad understanding of the principles of foods of plant origin and its classification</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to-

CO1. learn about various types of cereals, pulses and its types
CO2. know about fruits and vegetables, fats and oils, structure and types.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Plant based foods: Nutritional composition of plant based foods and their benefits. Plant-based foods vs. animal-based foods. Introduction to plant-based diets: vegetarianism, veganism, and flexitarianism Cereals - Structure and composition of cereals, Malting, gelatinization of starch, types of cereals. Rice- Structure and composition, parboiling of rice. **UNIT– II** Pulses- Structure and composition of pulses, toxic constituents in pulses. Processing of pulses- soaking, germination, decortications, cooking and fermentation **UNIT– III** Fats and Oils- Classification of lipids, types of fatty acids - saturated fatty acids, unsaturated fatty acids, essential fatty acids, trans fatty acids. Refining of oils - steam refining, alkali refining, bleaching, steam deodorization, hydrogenation. **UNIT-IV** Fruits and Vegetables- Classification of fruits and vegetables, general composition, enzymatic browning, names and sources of pigments. **Practical work** 1. To study gelatinization behavior of various starches 2. To study the concept of gluten formation of various flours. 3. To study malting and germination. 4. To study dextrinization in foods.
12	**Suggested Resources for Reading** 1. Food Science by Sunetra Roday., Oxford publication, Third edition, 2018

Semester IV

<table>
<tr><td>1</td><td>Course Title</td><td>Food Packaging Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 401</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
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</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To enable the students to understand about packaging and packaging materials, interaction of food items with packaging materials</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Comprehend the reason for food packaging.
CO2. Learn about different kinds of packaging
CO3. Understand about canning operations.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT I** Introduction to Food Packaging: Functions of packaging, Effect of environmental factors on quality of food. Estimation of shelf life, analysis of storage requirement, accelerated storage studies: Vacuum and Inert Gas Packaging: Tests on packaging materials, Mechanical strength (Tension, notch and tearing strengths), Gas and water vapor transmission rates. **UNIT II** Metal Cans as Packaging material: Types of Metallic Cans: Tin cans and Aluminum cans. Specialty of Open top sanitary cans, Lacquers and their use, Three piece cans and two piece cans, Aerosol Cans. Introduction to Canning operations: Can Reformer, Flanger, Seaming, Can closures, Glass jars and Bottles in food packaging, Design features and applications, Sterilization of jars and bottles. **UNIT III** Flexible Films Packaging: Formation of Films and pouches, Plastics used and their Specific applications. Rigid and Semi rigid plastic packaging methods. Copolymers and their applications.Co-extruded films and Laminates. Extrusion –Retort pouch packaging. Laminated Paper board Cartons, Fiber Board and Corrugated Card board packaging and their applications. **UNIT IV** Filling And Sealing Operations For Various Types of Packages: Closing and sealing of rigid plastic containers. Filling and sealing of flexible plastic containers, Seal types, hot wire sealing, hot bar sealing and impulse sealing. Active packaging, Moisture control, CO_2 and Oxygen scavenging, Modified atmosphere packaging: principles and applications. Permeability of gases in packs. **Practical work** **1.** To study quality and strength of packaging materials. **2.** Measurement of cartons' dimensions as per organizational standards. **3.** Measurement of dimensions of bottle mouth, cans and their caps. **4.** Determination of tensile strength of given material. **5.** Testing of chemical resistance of packaging materials. **6.** Demonstration of sealing processes used in food industry. **7.** Demonstration of filling process.
12	**Suggested Resources for Reading** 1. Coles, R., Dowell, D.M. & Kirwan, J, Food Packaging Technology, Black Well Publishing Ltd. 2. NIIR Board, Food Packaging Technology Handbook, National Institute of Industrial Research. 3. Pirenger, O.G. and Baver, A.L., Plastic Packaging Materials for Food, Wiley.

<table>
<tr><td>1</td><td>Course Title</td><td>Food Microbiology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 402</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
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(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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Theory Paper:
(4 X 2 marks) = 8 marks
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iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Students will study about the growth of different types of microorganisms based on various environmental factors and effect on food.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Learn about the different types of fermentation processes, equipments used and microbiological processes involved
CO2. Gain knowledge of significance and activities of microorganisms in food.
CO3. Know the microbial quality control and quality schemes used in food industries.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT-I** Introduction to food microbiology: Importance in food safety, quality control, and public health. Normal microbiological quality of common foods. Sources of microbes in food: Soil, air, water, plants, animals, food ingredients, equipment. Factors affecting microbial growth in foods: pH, water activity, temperature, humidity, storage conditions. **UNIT– II** Types of Fermentation process: Batch, continuous and fed batch (Principle and significance). Solid state and submerged fermentation, applications in food industry. Basics of fermenter: different parts and their function. **UNIT III** Fermented Food: Fermented vegetables, dairy products, meat, poultry and fish products. Fermented beverages: Cocoa, coffee, wine, beer and their health benefits. Concept of prebiotics, probiotics, synbiotics. Role in gut health and immunity **UNIT-IV** Microbial Food Spoilage: Factors responsible for food spoilage (internal and external factors), spoilage of specific food types : Milk and dairy products, Cereals, Fruits and vegetables. Spoilage microorganisms in raw and pasteurized foods, canned foods, soft drinks and fruit juices, refrigerated foods. Indicators of microbial food spoilage: visual, textural, sensory and chemical changes **Practical Work** 1. Isolation of bacteria and fungus from spoiled food. 2. Identification of isolated bacteria by simple, gram staining and negative staining 3. Identification of fungi by lactophenol cotton blue staining. 4. Methylene Blue reductase test.
12	**Suggested Resources for Reading** 1.Frazier WC and Westhoff D C and Vanitha NM (2017): Food Microbiology, 5th Ed. MaGraw Hill Education.. 2. Jay JM (2005): Modern Food Microbiology, 3rd Ed. CBS Publishers & Distributors. 3.Pelczar M, Chan ECS, Krieg N(2009): Microbiology : Application Based Approach, Tata McGraw Hill Education.

<table>
<tr><td>1</td><td>Course Title</td><td>Foods of Animal Origin</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 403</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Students will learn about processing of food from animal to handle its aspects of perishable nature.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Gain In-depth knowledge of food from animal sources.
CO2. Learn about production and processing techniques.
CO3. Understand of the concepts underlying Dairy processing as well as meat and poultry.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Nutritional Importance of Foods of Animal Origin, Role of animal foods in human nutrition, High-quality proteins and essential amino acids. Vitamins (B12, A, D) and minerals (iron, calcium, zinc) Comparison of animal vs plant protein, Categories of Foods of Animal Origin Overview and examples: Milk and dairy products, Meat and meat products, Poultry and eggs, Fish and seafood, Honey and other minor animal foods, Dairy Foods Different dairy animals: cow, buffalo, goat, sheep, camel. **UNIT– II** Meat: Definition of meat, Concept of carcass and dressing percentage. Classification of Meat: Red meat and white meat (characteristics and examples), Factors affecting meat colour Composition of Meat: Water, proteins, fats, minerals, vitamins Changes during post-mortem handling, Marbling and Meat Quality, Definition and importance of marbling Effect on tenderness, juiciness, and flavour Sources of Meat: Cattle, buffalo, sheep, goat, pig. Meat from wild animals (brief overview) Nutritive Value of Meat, Biological value of meat proteins. **UNIT– III** Composition and Nutritive Value of Egg. Proteins, lipids, vitamins, minerals Egg as a "complete food", Egg Proteins: Ovalbumin, ovotransferrin, ovomucin, lysozyme, Functional properties of egg proteins Egg Quality and Freshness, Characteristics of fresh egg, Factors affecting egg quality, Deterioration of egg quality during storage. Broilers and Layers **UNIT-IV** Fish as Food, Importance of fish in human diet Lean and fatty fish Classification of Fish: Freshwater fish and Marine fish Aquaculture: Definition and scope, Major cultured fish species. Advantages of aquaculture Composition of Fish: Proteins, fats (omega-3 fatty acids), minerals, vitamins Freshness Characteristics of Fish: Physical, sensory, and chemical indicators Spoilage of Fish: Microbiological spoilage, Physiological spoilage, Biochemical spoilage. **Practical work** 1. Prepare a report on any food obtained from animals in detail from procurement till processing. 2. Determination of egg components (shell strength, egg weight and yolk index). 3. Grading and quality evaluation of egg.
12	**Suggested Resources for Reading** 1.Bawa. A.S, O.P Chauhan et al. Food Science. New India Publishing agency, 2013 2.Roday,S. Food Science, Oxford publication, 2011. Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Documentation and Record Keeping in food industry</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 404</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To enable the students to understand the importance of documentation in food industry and study preparation of records.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand documentation and understand preparation of records.
CO2. Prepare different types of documents as per requirement.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Importance of documents and record keeping in food industry. Types of documents: policies, manuals, procedures, records, reports. Principles of good documentation practices (GDP). Documentation flow in a food processing unit. Common documentation errors and their consequences. **UNIT– II** HACCP documentation: Hazard analysis sheets, CCP monitoring records, Retention of records, Review of records, disposal and retrieval of HACCP records. Documentation under: Good Manufacturing Practices (GMP), Good Hygiene Practices (GHP) **UNIT– III** Standard Operating Procedures (SOPs) for: Raw material handling. Processing and preservation operations. Equipment operation and maintenance logs. Cleaning and sanitation records (CIP, SSOP). Internal and external audit documentation **UNIT-IV** Data Protection: Confidentiality maintenance of records, Concerns related to data theft in food industry, Role of FDA and FSMA in protection of available food data, Role of ICT in data protection. General laws for record maintenance and protection. **Practical work** 1. Preparation of SOP for fruit/cereal processing operation. 2. Design of batch manufacturing record format. 3. Prepare of HACCP monitoring record for food product. 4. Prepare of audit checklist for a small food processing unit.
12	**Suggested Resources for Reading** **1.** Pierson, M. D., Corlett, D.A., HACCP: Principles and Applications, Springer. Note: Latest edition of text books may be used.

Semester V

<table>
<tr><td>1</td><td>Course Title</td><td>Food Analysis</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 501</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits (2+2)</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

The exam of this course will be of 3 hours duration.
There will be no objective type questions.
Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
All questions carry equal marks
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Determine moisture, protein, fat, ash content, and understand interactions of food components and nutrient value.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Understand working principles and application of various food analysis techniques
CO2. Learn methods to analyze physical, chemical, and rheological properties of foods.
CO3. Use analytical instruments to determine specific compounds in food.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

12	**Course Content** **UNIT– I** Food analysis: Types of food analysis (compositional, nutritional, and safety-related). Sampling: Sample collection, handling, storage, and preparation. Good Laboratory Practices (GLP). Sensory evaluation of foods; Difference, rating and sensitivity tests, types of sensory panels, criteria for panelist selection, sensory laboratory layout and design. **UNIT– II** Proximate analysis of food: Role of proximate analysis in nutritional evaluation. Principles and standard analytical methods for the determination of moisture content, ash content, crude fat, crude protein, total carbohydrates, and crude fiber in foods **UNIT– III** Colorimetery: Principles, types and applications in food analysis. Centrifugation: Principles and applications of Density gradient, Differential centrifugation and ultracentrifugation. X-ray analysis of foods: Principles of X-ray techniques; applications in detection of foreign materials, structural analysis. **UNIT-IV** Microscopy: Bright field microscopy, Fluorescent microscopy, Dark field microscopy. Scanning electron microscope (SEM), Transmission electron microscope (TEM). Rapid methods of microbial analysis in food **Practical Work** 1. To find moisture content of given food sample by hot air oven method 2. To estimate the amount of protein content by the Kjeldahl method. 3. To find out ash content in the given food sample. 4. To find out the amount of crude fibre in a given food sample. 5. Clarification of fruit juice using centrifugation. 6. Preparation and Evaluation of Sensory Scorecards.
13	**Suggested Resources for Reading** 1.. Ghoshal, Fundamentals of Bioanalytical Techniques and Instrumentation (2013), CBS Publishers. 2. Bioanalytical Techniques (2010) M. L. Srivastava Alpha Science Intl Ltd. Food Analysis by Nielsen, Suzanne (Ed.) -Springer, 2010. 3. Methods in Food Analysis by Rui M. S. Cruz, Igor Khmelinskii, Margarida Vieira -CRC Press, 2014. 4. Handbook of Food Analysis by Leo M.L. Nollet, Fidel Toldra- CRC Press, 2015.

<table>
<tr><td>1</td><td>Course Title</td><td>Industrial safety</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 502</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

The exam of this course will be of 3 hours duration.
There will be no objective type questions.
Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
All questions carry equal marks
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Understand the industrial laws, regulations and source models.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Illustrate the importance of food quality standards and of Quality Assurance.
CO2. Apply the methods of prevention of fire and explosions.
CO3.Understand the methods of hazard identification and preventive measures.
CO4. Conduct the Accident/incident analysis with different techniques</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

12	**Course Content** **UNIT– I** Concept and importance of industrial safety. Types of hazards: physical, chemical, biological, mechanical, electrical. Common accidents in food processing and preservation plants. Safety signs, symbols, and color codes. Safety responsibilities of management and worker **UNIT– II** Hazards in fruit, vegetable, dairy, meat, and cereal processing units. Safety in cutting, slicing, grinding, blanching, cooking, drying, freezing, and canning. Safety in use of boilers, steam systems, autoclaves, and pressure vessels. Refrigeration and cold storage safety. Handling of food-grade chemicals, preservatives, and cleaning agents. Machine guarding. Lock out and tag out system. **UNIT– III** Industrial hygiene and workplace cleanliness. Personal hygiene and GMPs in food industries. Cleaning and sanitation (CIP & SIP systems). Protective clothing and personal protective equipment (PPE). Prevention of contamination and cross-contamination. Waste handling and effluent management safety. **UNIT-IV** Fire hazards in food processing plants. Types of fires and fire-fighting equipments. Electrical safety and grounding systems. Occupational health hazards: heat stress, noise, dust, chemicals. First aid and emergency response. Safety audits, risk assessment, and hazard analysis. Overview of OSHA concepts.
13	**Suggested Resources for Reading** 1. Chemical process safety fundamentals with applications by Crawl D.A. And Louver J.A.Prentice Hall, 2000. 2. Safety Health and Environmental Protection by Wentz, C.A.-McGraw Hill, 2001. 3. https://www.osha.gov

<table>
<tr><td>1</td><td>Course Title</td><td>Principles of Management in Food Industry</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 503</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>60 hours of lectures for Theory
(1 credit = 15 hours of theory learning based on 4hrs/week)
(1 Semester=15 weeks)</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To make the students understand the concepts of management & their application in the food industry.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to-
CO1. understand the concept of management in industries
CO2. develop leadership and motivation skills required in industry</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Organization: Concept of organization, Goals and policies Formal & Informal, Centralized & Decentralized organizations. Management: Concept of Management. Types of management. Levels of management. . External & internal factors affecting management. **UNIT– II** Planning: Definition, Nature and importance of planning in food processing. Advantages and limitations. Steps in planning. Planning assumptions (market demand, raw material and shelf life considerations). Decision-making: Types of decisions. Step by step decision making process. Coordination and Controlling: Coordination: Concept, need and techniques. Controlling: Objectives and Process of control. Devices of control. **UNIT– III** Communication: Definition, Nature, Types of communication, Barriers to communication, Methods of improving communication and effectiveness. Staffing: Recruitment and selection basis, SWOT analysis of staff. Recognizing and rewarding staff members. **UNIT-IV** Leadership: Definition. Different styles of leadership. Characteristics of a good leader. Motivation: Definition. Nature & importance. Benefits of motivated staff.
12	**Suggested Resources for Reading** 1. Psychology for managers by Shashi K. Gupta and Rosy Joshi -Kalyani Publishers, 2007. 2. Principles of Management by T. Ramasamy -Himalaya Publishing house, 2010. 3. Principles of Management by Shikha Taneja -S K Kataria & Sons Publication, 2012.

<table>
<tr><td>1</td><td>Course Title</td><td>Technology Of Fruits And Vegetables</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 504</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits (2+2)</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

The exam of this course will be of 3 hours duration.
There will be no objective type questions.
Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
All questions carry equal marks
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Students will be able to Prepare, dry and storage fruits and vegetables with appropriate methods.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Prepare, preserve and store jam, jelly and marmalades by using appropriate machines.
CO2. Prepare and package fruit beverages such as Squashes, RTS, Nectar, Cordial, Crush and Syrup by using appropriate machines.
CO3. Prepare and preserve Tomato products by using appropriate machines.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

12	**Course Content** **UNIT– I** Classification of fruit and vegetable (morphology and processing suitability). Maturity and ripeness indices for major fruit and vegetables. Harvesting methods (washing, grading, waxing and pre-cooling), Post harvest losses: Mechanical, physiological, pathological, biochemical causes and prevention. Storage and transportation of fruits and vegetables. **UNIT– II** Processing of fruits: Fruit Juices (extraction, clarification, preservation and packaging), nectars, squashes, syrups, Concentrated fruit products and pulps. Jam, Jellies, Marmalades. Dehydrated fruit products and powders. Quality defects in fruit products. **UNIT– III** Processing of vegetables: Canning of vegetables (Principles and processing steps), blanching methods and importance. Pickling: Oil, vinegar and brine pickles. Chutneys. Dehydrated vegetable products. Tomato products: Selection of tomatoes, pulping and processing of tomato juice, Tomato puree, ketchup, sauce. **UNIT-IV** Packaging: Materials and methods for fruit and vegetables. Spoilage and defects in processed products. Utilization of fruit and vegetable waste (peels, pomace, seeds). Value addition and Entrepreneurship opportunities. **Practical work** 1. Estimation of total soluble solids (TSS) and acidity ratio in fruits 2. Preparation and evaluation of tomato products. 3. Preparation of fruit juice and squash. 4. Dehydration and Rehydration of fruits and vegetables. 5. Drying characteristics of fruits: Sun versus mechanical drying. 6. Shelf life evaluation of processed products.
13	**Suggested Resources for Reading** 1. Food Science- Potter, CBS publishers. 2. Food Processing and Preservation- Subbulaksmi G., and Udipi S 3. Principles of Food Science, Vol. II- G. Borgstron, Mc. Millan Co. Ltd. London 4. Principles of food preservation Part I& II- Owen R. Fenemma.

<table>
<tr><td>1</td><td>Course Title</td><td>Technology of Plantation Crops and Oilseeds</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 505</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits (2+2)</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

The exam of this course will be of 3 hours duration.
There will be no objective type questions.
Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
All questions carry equal marks
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Students will learn about Principles of Dairy, processing of technologies to handle physical aspects of fluid based milk and its various products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Knowledge of harvesting, processing, and creating value-added products
CO2. To understand the processing of plantation crops.
CO3.Gain knowledge about processing of coffee and cocoa processes and byproducts.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

12	Course Content **UNIT– I** Plantation Crops: Importance and classification (tea, coffee, cocoa, coconut, spices). Chemical composition and nutritive value. Quality parameters and grading standards. Oilseeds: Major oilseeds (groundnut, mustard, soybean, sunflower, sesame, cottonseed) General structure and chemical composition of oilseeds. Factors affecting oil quality **UNIT– II** Tea Processing: Green, black and oolong tea processing. Withering, rolling, fermentation, firing and grading. Quality evaluation of tea Coffee Processing: Wet and dry processing methods. Roasting, grinding and packaging. Coffee quality and defects Cocoa Processing: Cocoa fermentation, drying and chocolate manufacture. Coconut processing: Copra, coconut oil, desiccated coconut. **UNIT– III** Processing of Oilseeds and Edible Oils: Cleaning, grading, dehulling and conditioning. Oil extraction methods: mechanical expelling, solvent extraction Crude oil processing and refining. Degumming, neutralization, bleaching and deodorization Hydrogenation and interesterification. By-products: oil cake and meal **UNIT-IV** Rancidity: types, causes and prevention. Storage methods of oilseeds and edible oils. Packaging materials for oils and plantation crop products. Quality analysis of oils (acid value, iodine value, peroxide value). Overview of AGMARK and BIS for plantation crops and oilseeds. **Practical work** 1) Determination of refractive index and specific gravity of fats and oils. 2) Determination of smoke point and percent fat absorption for different fat and oils. 3) Determination of percent free fatty acids. 4) Extraction and estimation of polyphenol content in tea and coffee. 5) Determine iodine value, acid value and rancidity. 6) Coffee roasting and grinding.
13	**Suggested Resources for Reading** 1. Tea Production and Processing. B. Banerjee, Oxford & IBH Pub. Co., 1st Edition, 1993. 2. Coffee Technology. M. Sivetz, AVI publishing Co., 1st Edition, 1979. Note: Latest edition of text books may be used.

Semester VI

1	**Course Title**	**Bioanalytical Techniques**
2	**Course Code**	BFP 601
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits (2+2)

<table>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
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ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To enable the students to understand the properties of food and detection of microorganisms using various basic and advance tools and techniques.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. explain and compare the principles and applications of major analytical techniques.
CO2. apply standard laboratory procedures to analyze food samples.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Chromatography: Paper and Thin Layer Chromatography: Principle and applications. Gel filtration, Ion exchange chromatography, HPLC, Gas chromatogtaphy: Principle and applications. **UNIT– II** Electrophoresis: Principle and applications in detection of food constituents and contaminants in food items. Biosensors: Components and working principle (bioreceptor, transducer, signal processor) Applications of biosensors: detection of pathogens, toxins, allergens, adulterants. **UNIT– III** UV-Vis spectroscopy, Liquid chromatography-Mass spectrometry (LC-MS), Fourier Transform Infrared Spectroscopy (FTIR), Nuclear Magnetic Resonance (NMR), ELISA, PCR and RT-PCR: Principle and applications **UNIT-IV** Thermogravimetric Analysis, Differential Scanning Calorimetry, Differential Thermal analysis, Melting, crystallization, glass transition in fats, chocolates. AI for Food Analysis: Predicting spoilage, quality grading, adulteration detection. **Practical Work** To estimate amount of protein content by Lowry method. To perform estimation of total carbohydrates in given food sample. To find out the amount of crude fibre in given food sample. 4.To test different food samples for adulteration: milk, ghee, butter, honey.
12	**Suggested Resources for Reading** Food Analysis by Nielsen, Suzanne (Ed.) -Springer, 2010. Methods in Food Analysis by Rui M. S. Cruz, Igor Khmelinskii, Margarida Vieira -CRC Press, 2014. 3. Handbook of Food Analysis by Leo M.L. Nollet, Fidel Toldra- CRC Press, 2015. 4. Chemical Analysis of Food: Techniques and Applications by Yolanda Picó -Elsevier 2012.

<table>
<tr><td>1</td><td>Course</td><td>Food Safety, Standards And Regulations</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 602</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
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iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To make the students conversant with latest rules and regulation in Food safety as per FSSAI guidelines especially dealing with work place.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. evaluate national food safety regulations by explaining the key provisions, hygienic and sanitary requirements, allowable contaminant limits, and regulatory mandates outlined under the Food Safety and Standards Act, 2006 and subsequent FSSAI regulations from 2011.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Food safety: Importance of food safety, concerns related to food process hygiene and food products. Hygiene indicator organisms for various foods. Overview of Food Safety and Standards Act, 2006: Salient features of Food Safety & Standards Act, 2006. Contaminants and food safety: General metal contaminants and their effects in food products lead, copper, arsenic, tin, zinc, mercury. Pesticides and insecticides as food contaminants. General regulations specified for different contaminants in food in India. **UNIT– II** Food safety and Standards (Licensing and Registration of Food businesses) regulation, 2011: Salient feature of schedule 1, 2, 3 Salient feature of schedule 4: General Hygienic and Sanitary practices to be followed by all Food Business operators Specific Hygienic and Sanitary Practices to be followed by Food Business Operators engaged in manufacture, processing, storing and selling of Milk and Milk Products **UNIT– III** Food Safety and Standards (Food product standards and Food Additives) Regulation, 2011 (part I): Salient features of regulation. First amendment regulation 2025. Food Safety and Standards (Food product standards and food additives) Regulation, 2011 (part II): Salient features of Regulation **UNIT-IV** Food Safety and Standards (Prohibition and Restriction on sales) Regulation, 2011: Salient features of regulation Food Safety and Standards (Packaging and Labeling) Regulation, 2011: Salient features of regulation Food Safety and Standards (contaminants, toxins and residues) Regulation, 2011: Salient features of regulation Food Safety and Standards (Laboratory and sampling analysis) Regulation, 2011: Salient features of regulation **Practical work** The student will study and prepare a report of Food Safety Standards and Regulation as per FSSAI pertaining to workplace especially health hazards, hygiene practices and disposal of waste etc.
12	**Suggested Resources for Reading** 1. A Practical Guide to Food Laws and Regulations by Kiron Prabhakar-Bloomsbury India, 2016. 2. Food Safety and Standards Authority of India: http://www.fssai.gov.in/home/fsslegislation/fss - regulations.html 3. http://fssai.gov.in/upload/uploadfiles/files/Palm%20Oil_FPSFA_Notification.pdf

<table>
<tr><td>1</td><td>Course Title</td><td>Food Pathogens</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 603</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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<tr><td>6</td><td>Max. Marks and Time</td><td>
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
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iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To create awareness about the food pathogens , sources of food and water borne diseases, their causative agents, characteristics, pathogenesis, Isolation of food borne pathogens and their detection.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to-
CO1. analyze the sources, epidemiology, and mechanisms of food and water-borne diseases
CO2. apply laboratory and preventive strategies to detect, differentiate, and control foodborne pathogens and mycotoxins by using standard microbiological techniques</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Food and water borne diseases: Introduction, sources of food contamination, Significance and incidence of foodborne illness. Risk factors associated with food borne illness. Common site of food borne illness. Pathogenesis of the diarrhoeal disease and its epidemiology. **UNIT– II** Bacterial agents of food borne illness: Characteristics, pathogenesis, clinical symptoms, isolation and detection methods of *Bacillus* sp., *Clostridium botulinum Escherichia coli*, *Listeria monocytogenes*, *Salmonella* sp., *Shigella* sp., *Staphylococcus aureus*, **UNIT– III** Foodborne Virus: Characteristics, pathogenesis, clinical symptoms, isolation and detection methods of Polio virus, Rota virus and Noro virus, Hepatitis A and Hepatitis C. Protozoan agents for food borne illness: Pathogenesis and prevention of *Giardia lamblia* and *Entamoeba histolytica*. **UNIT-IV** Fungi and mycotoxins: Mycotoxins of *Aspergillus* sp., *Penicillium* sp., and their association with various foods. Control of mycotoxins in food. Methods to control foodborne disease outbreaks in Industrial and food service establishment/household specific control measure. **Practical work** 1. Analysis of mycotoxins in fungal-contaminated food materials. 2. Presumptive test for coliforms in water or food samples 3. Differential test of Staphylococci through growth on agar plates. 4. Differentiation and identification of Streptococci. 5. Control of growth of pathogens in food samples.
12	**Suggested Resources for Reading** Foodborne Pathogens and Food Safety by Md. Latiful Bari, Dike O. Ukuku-CRC Press, 2015. 2. Foodborne Pathogens: Hazards, Risk Analysis and Control by Clive de Blackburn Peter McClure -Elsevier, 2009.

<table>
<tr><td>1</td><td>Course Title</td><td>Waste Management and Environmental Control in Food Processing</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 604</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits (2+2)</td></tr>
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

The exam of this course will be of 3 hours duration.
There will be no objective type questions.
Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
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All questions carry equal marks
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The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand sources and types of waste generated in food processing industries.
To impart knowledge of waste treatment technologies and disposal methods.
To introduce sustainable waste management and resource recovery.
To create awareness about environmental laws and regulatory compliance.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Identify solid, liquid, and gaseous wastes in food processing plants.
CO2. Explain treatment technologies for food industry wastes
CO3. Apply waste minimization and reuse strategies.
CO4. Understand environmental regulations and compliance</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Definition and importance of waste management. Characteristics of food industry wastes Types of waste: Solid waste (peels, pomace, husk, shells), Liquid waste (effluents, wash water), Gaseous waste (odours, emissions). Environmental impact of food processing wastes **UNIT– II** Segregation of waste at source. Handling, storage and transportation of waste. Water conservation and reuse in food plants. Good housekeeping practices. Role of GMP and GHP in waste reduction Occupational safety during waste handling. **UNIT– III** Waste treatment: Disposal methods: landfilling, incineration. Liquid waste treatment: Primary treatment (screening, sedimentation), Secondary treatment (Aerobic treatment (activated sludge, trickling filters). Anaerobic treatment (UASB, anaerobic digesters). Tertiary treatment and effluent reuse. Solid Waste Treatment: Composting of organic waste, Vermicomposting, biogas production, animal feed applications. Gaseous Waste Management: Odour control techniques. **UNIT-IV** Waste conversion: Biofertilizers, Bioenergy, Food ingredients (pectin, fiber, antioxidants). Environmental impact assessment (EIA). Environmental laws and standards: Salient features of Environment Protection Act, 1986, Water (Prevention and Control of Pollution) Act 1974, Air (Prevention and Control of Pollution) Act 1981. Role of Central Pollution control board in waste management. Overview of sustainability and circular economy in food industries. **Practical work** 1. Identification and classification of waste from a food processing unit. 2. Measurement of pH, BOD and COD of food industry effluent (demonstration). 3. Study of solid waste composting methods. 4. Case study on waste management in fruit/vegetable processing industry. 5. Preparation of waste flow diagram for a food processing plant. 6. Visit to Effluent Treatment Plant (ETP) / virtual tour. 7. Preparation of a waste management plan for a small food unit.
12	**Suggested Resources for Reading** 1. Kapoor, B., Singh, R., Kapoor, D., & Gautam, V. (Eds.). (2022). Environmental sustainability in the food industry: a green perspective. CRC Press. 2. Panda, H. (2011). The Complete Book on Managing Food Processing Industry Waste. Asia Pacific Business Press Inc. 3. Arvanitoyannis, I. S. (2010). Waste management for the food industries. Academic Press.

<table>
<tr><td></td><td>Course Title</td><td>Meat & Seafood processing</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 605</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits (2+2)</td></tr>
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

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ii. There will be no objective type questions.
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iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To create awareness about the processing of meat and seafood/fish and their storage and handling techniques</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to-

CO1. learn about various types of cereals, pulses and its types
CO2. know about fruits and vegetables, fats and oils, structure and types.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Overview of Meat Industry in India: Current status and economic importance. Meat: Sources (animals and fish), Composition and nutritive and value, Slaughtering and dressing of meat animal; Quality evaluation of eviscerated carcasses; post mortem changes in meat: Rigor mortis, tenderization, pH changes, water-holding capacity **UNIT– II** Meat Processing: Primary and Secondary Processing, Cutting, deboning, mincing, curing, smoking, marination Tenderization methods: mechanical, enzymatic, chemical . **UNIT– III** Fish: Factors affecting quality of fresh fish, fish dressing, chilling, freezing, salting and canning and glazing of fish. Microbial spoilage of meat and fish, control measures, **UNIT-IV** Preservation techniques: Natural preservatives, Thermal processing, dehydration, curing, freezing, irradiation Restructured meat products, meat analogs, value added products (fish fingers, surimi) **Practical work** Identification of meat and fish cuts using charts Calculation based exercise on moisture, fat and protein estimate Thermal processing calculation (F0 value, lethality and cold spots) Microbial spoilage pattern of meat and fish (case studies/analysis) Study of packaging systems for meat and seafood Regulatory standards for meat and seafood Visit to meat industry and report preparation
12	**Suggested Resources for Reading** Hui, Y. H. (Ed.). (2012). Handbook of meat and meat processing. CRC press. Toldrá, F. (Ed.). (2010). Handbook of meat processing. John Wiley & Sons.

<table>
<tr><td>1</td><td>Course Title</td><td>Egg & Poultry processing</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 606</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits (2+2)</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
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Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
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iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

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iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Safety Analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To provide knowledge on product development, quality evaluation, safety concerns, and modern industrial practices in poultry and egg-based food production.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to-
CO1. learn about the structure, composition, nutritional significance, processing technologies, and quality determinants of eggs and poultry products
CO2. perform practical processing and design HACCP models used in egg and poultry product handling.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Poultry industry in India. Importance of poultry and eggs in human nutrition. Broilers, layers, indigenous birds in poultry industry. Types & forms of feeds and feeding methods, feeding equipments. HACCP in egg and poultry processing plants. FSSAI standards for poultry meat and egg products. . **UNIT– II** **UNIT– III** Liquid egg processing: Steps in processing of whole egg, yolk, albumin separation, Drying of eggs: spray drying and freeze drying Egg-based products: mayonnaise, custards, bakery applications **UNIT-IV** Egg quality deterioration: Chemical changes, green rings, off-odors, microbial spoilage Preservation Techniques: egg powder manufacture, egg coatings, **Practical work** Egg grading (weight, shape index, candling) Detection of egg and poultry meat spoilage To study the shelf-life of eggs Preparing a generic HACCP model for poultry./egg processing plants Visit to poultry industry.
12	**Suggested Resources for Reading** Stadelman, W.J., Newkirk, D., & Newby, L. (1995). Egg Science and Technology (4th ed.). CRC Press. https://doi.org/10.1201/9780203758878 Poultry, Egg and Fish Proceesing Technology by Navneen Kumar 2018. Guerrero-Legarreta, I., & Hui, Y. H. (2010). Handbook of Poultry Science and Technology. John Wiley & Sons, Inc.